CURRICULUM SPECIFICATIONS
CHEMISTRY 1 (EC015)

1.	Name and code of course:	Chemistry 1 (EC015)
2.	Synopsis:	This course is designed to enable students to learn the basic concepts and principles of chemistry. This course covers matter, atomic structures, periodic table, chemical bonding, states of matter, chemical equilibrium and ionic equilibria.
3.	Name(s) of academic staff:	Siti Warda binti Selamat
4.	Semester offered:	Semester I
5.	Credit value:	5
6.	Prerequisite/ co-requisite (if any):	None
7.	Course Learning Outcomes (CLO): At the end of the course, students should be able to: 1. Explain the fundamental concepts in chemistry. (C2, PLO 1, MQF LOC i) 2. Solve problems with analytical and critical thinking using chemistry facts and principles. (C4, PLO 2, MQF LOC ii) 3. Apply the appropriate scientific laboratory skills in chemistry experiments. (P3, PLO 3, MQF LOC iii a) 4. Analyse numerical problems in chemistry. (C4, PLO 7, MQF LOC iii e) *Note*: *The* ***MQF* 2.0 *learning outcomes clusters* (*LOC*)** *are*: *i. Knowledge and understanding* *ii. Cognitive skills* *iii. Functional work skills with focus on:* *a. Practical skills* *b. Interpersonal skills* *c. Communication skills* *d. Digital skills* *e. Numeracy skills* *f. Leadership, autonomy and responsibility* *iv. Personal and entrepreneurial skills* *v. Ethics and professionalism.*	

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8. Mapping of Programme Learning Objectives (PLO) to Programme Educational Outcomes (PEO):

PEO / PLO	PEO 1	PEO 2	PEO 3	PEO 4	PEO 5
PLO 1	✓				
PLO 2		✓			
PLO 3	✓				
PLO 4			✓		
PLO 5			✓		
PLO 6				✓	
PLO 7		✓			
PLO 8					✓
PLO 9				✓	
PLO 10					✓

Programme Educational Objectives (PEO)

Upon a year of graduation from the programme, graduates are:

PEO 1 Knowledgeable and technically competent in engineering disciplines in-line with higher educational institution requirement.
PEO 2 Able to apply information and use data to solve problems in engineering disciplines.
PEO 3 Able to communicate competently and collaborate effectively in group work to compete in higher education environment.
PEO 4 Able to use basic information technologies and engage in life-long learning to continue the acquisition of new knowledge and skills.
PEO 5 Able to demonstrate leadership skills and practice good values and ethics in managing organisations.

Programme Learning Outcomes (PLO)

At the end of the programme, students should be able to:

PLO 1 Acquire basic knowledge of engineering, science and mathematics for higher education purposes.
PLO 2 Apply critical thinking and problem solving skills in engineering, science and mathematics.
PLO 3 Demonstrate manipulative skills in workshop and laboratory works.
PLO 4 Collaborate in group work with skills required for higher education purposes.
PLO 5 Deliver ideas, information, problems and solution in verbal and written communication.
PLO 6 Use basic digital technology to seek and analyse data for management of information.
PLO 7 Interpret familiar and uncomplicated numerical data to solve problems.
PLO 8 Demonstrate leadership, autonomy and responsibility in managing organization.
PLO 9 Initiate self-improvement through independent learning.
PLO 10 Practice good values, attitude, ethics and accountability in STEM and professionalism.

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CURRICULUM SPECIFICATIONS
CHEMISTRY 1 (EC015)

9. Mapping of Course Learning Outcomes (CLO) to Programme Learning Outcomes (PLO), Teaching Methods and Assessment Methods:

PLO / CLO	PLO 1	PLO 2	PLO 3	PLO 4	PLO 5	PLO 6	PLO 7	PLO 8	PLO 9	PLO 10	Teaching Method	Assessment Method
CLO 1	✓										Lecture	Summative Assessment Test
CLO 2		✓									Tutorial	Examination Assignment Lab Report
CLO 3			✓								Practical	Practical Test
CLO 4							✓				Tutorial	Assignment

(This description must be read together with Standard 2.1.2, 2.2.1 and 2.2.2 in Area 2)

10. Transferable Skills (if applicable): Critical Thinking and Problem Solving
Numeracy Skill

11. Distribution of Student Learning Time (SLT):

Course Content Outline	CLO	Teaching & Learning Activities						Total SLT
		Guided Learning, e-Learning (F2F)				Guided Learning (NF2F) e.g. e-Learning	Independent Learning (NF2F)	
		L	T	P	O			
1.0 Matter 1.1 Atoms and molecules 1.2 Mole concept 1.3 Stoichiometry	1, 2, 3	2	18	6	0	0	20	46
2.0 Atomic Structure 2.1 Bohr's atomic model 2.2 Quantum mechanics 2.3 Electronic configuration	1,2	3	6	0	0	0	9	18
3.0 Periodic Table 3.1 Classification of elements 3.2 Periodicity	1,2,4	2	8	0	0	0	10	20
4.0 Chemical Bonding 4.1 Lewis structure 4.2 Molecular shape and polarity 4.3 Orbital overlap and hybridisation 4.4 Intermolecular forces 4.5 Metallic bond	1,2	3	10	0	0	0	13	26

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	5.0 States of Matter 5.1 Gas 5.2 Liquids 5.3 Solids 5.4 Phase diagram	1, 2, 3	3	7	2	0	0	10	22
	6.0 Chemical Equilibrium 6.1 Dynamic equilibrium 6.2 Equilibrium constants 6.3 Le Chatelier's Principle	1, 2, 3	2	7	2	0	0	9	20
	7.0 Ionic Equilibria 7.1 Acids and bases 7.2 Acid–base titrations 7.3 Solubility equilibria	1, 2, 3	3	10	2	0	0	13	28
	Total		**18**	**66**	**12**	**0**	**0**	**84**	**180**

Continuous Assessment	**Percentage (%)**	**F2F**	**NF2F**	**Total SLT**
1. Assignment (Topic 3 – CTPS)	10	0	2	2
2. Assignment (Topic 3 – Numeracy)	5	0	1	1
3. Practical Test	10	1	3	4
4. Lab Report	15	1	3	4
Final Assessment	**Percentage (%)**	**F2F**	**NF2F**	**Total SLT**
1. Summative Assessment Test	20	1.5	4.5	6
2. Final Examination	40	2	6	8
GRAND TOTAL SLT				**205**

L = Lecture, T = Tutorial, P = Practical, O = Others, F2F = Face-to-Face, NF2F = Non Face-to-Face

12.	Special requirement or resources:	None
3.	References:	Brown, T. E., LeMay, H. E., Bursten, B. E., Murphy, C., Woodward, P. & Stoltzfus, M. E. (2018). *Chemistry: The Central Science* (14th ed.). Pearson Education. Silberberg, M. & Amateis, P. (2021). *Chemistry: The Molecular Nature of Matter and Change* (9th ed.). McGraw-Hill. Chang, R. & Overby, J. (2019). *Chemistry* (13th ed.). McGraw-Hill.

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14.	Other additional information:	Teaching Learning Outcomes:

Topic 1: Matter

1.1 Atoms and molecules
- a) Write isotopic notation.
- b) Interpret mass spectrum.
- c) Calculate the average atomic mass of an element given the relative abundances of isotopes or a mass spectrum.
  **include calculation on relative atomic mass based on C-12.*

1.2 Mole concept
- a) Define the terms empirical and molecular formulae.
- b) Determine empirical and molecular formulae from mass composition or combustion data.
- c) Determine the empirical formula (formula unit) from experiment.
  (Experiment 1: Determination of a formula unit of a compound).
- d) Define each of the following concentration measurements:
  - i. molarity (M);
  - ii. molality (m);
  - iii. mole fraction (X);
  - iv. percentage by mass (%w/w); and
  - v. percentage by volume (%v/v).
- e) Calculate each of the following concentration measurements:
  - i. molarity (M);
  - ii. molality (m);
  - iii. mole fraction (X);
  - iv. percentage by mass (%w/w); and
  - v. percentage by volume (%v/v).

1.3 Stoichiometry
- a) Write a balanced chemical equation:
  - i. by inspection method; and
  - ii. by ion-electron method (redox equation).
- b) Define:
  - i. limiting reactant; and
  - ii. percentage yield.
- c) Perform stoichiometric calculations using mole concept including limiting reactant and percentage yield.
  (Experiment 2: Acid-Base Titration - Determination of the Concentration of Hydrochloric Acid)
  (Experiment 3: Determination of Molar Mass of a Metal)

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Topic 2: Atomic Structure

2.1 Bohr's atomic model

a) Describe Bohr's atomic model.
b) Explain the existence of energy levels in an atom.
c) Calculate the energy of an electron using:

$$E_n = -R_H\left(\frac{1}{n^2}\right), \text{ where } R_H = 2.18\times10^{-18}\ J$$

d) Describe the formation of line spectrum of hydrogen atom.
e) Illustrate the formation of Lyman, Balmer, Paschen, Brackett and Pfund series.
f) Calculate the energy change of an electron during transition using:

$$\Delta E = R_H\left(\frac{1}{n_i^2} - \frac{1}{n_f^2}\right), \text{ where } R_H = 2.18\times10^{-18}\ J$$

g) Calculate the energy of photon emitted by an electron that produces a particular wavelength during transition using:

$$\Delta E = h\nu, \text{ where } \nu = \frac{c}{\lambda}$$

h) Perform calculations involving the Rydberg equation:

$$\frac{1}{\lambda} = R_H\left(\frac{1}{n_1^2} - \frac{1}{n_2^2}\right), \text{ where } R_H = 1.097\times10^{7}\ m^{-1} \text{ and } n_1 < n_2$$

i) Calculate ionisation energy of hydrogen atom from Lyman series.
j) State the limitation of Bohr's atomic model.
k) State the dual nature of electron using de Broglie's postulate and Heisenberg's uncertainty principle.

2.2 Quantum mechanics

a) Define the term orbital.
b) Explain all four quantum numbers of an electron in an orbital:
   i. principal quantum number, n
   ii. angular momentum quantum number, ℓ
   iii. magnetic quantum number, m
   iv. electron spin quantum number, s
c) Sketch the 3-D shapes of s, p and d orbitals.

2.3 Electronic configuration

a) Explain Aufbau Principle, Hund's Rule and Pauli Exclusion Principle.
b) Predict the electronic configuration of atoms and monoatomic ions using *spdf* notation and orbital diagram.
c) Explain the anomalous electronic configurations of copper and chromium.

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Topic 3: Periodic Table

3.1 Classification of elements
- a) Describe period, group and block (*s,p,d,f*).
- b) Deduce the position of elements in the periodic table from its electronics configuration.

3.2 Periodicity
- a) Describe the variation in atomic radii:
  - i. across a period;
  - ii. across the first row of transition element; and
  - iii. down a group.
- b) Analyse the variation in atomic radii:
  - i. across a period;
  - ii. across the first row of transition element; and
  - iii. down a group.
- c) Compare the atomic radius of an element and its corresponding ionic radius.
- d) Define the term isoelectronic.
- e) Compare the radius of isoelectronic species.
- f) Analyse the variation in the ionic radii across period 2 and period 3.
- g) Define the first and second ionisation energies.
- h) Analyse the variations in the first ionisation energy across a period and down a group.
- i) Explain the increase in the successive ionisation energies of an element.
- j) Deduce the electronic configuration of an element and its position in the periodic table based on successive ionisation energy data.
- k) Define electronegativity.
- l) Explain the variation in electronegativity of elements.
- m) Explain the acid-base character of oxides of elements in Period 3.

**relate the acid-base character to the types of bonding (ionic or covalent).*

Topic 4: Chemical Bonding

4.1 Lewis structure
- a) State the octet rule.
- b) Describe how atoms achieve stability by attaining stable configuration of:
  - i. noble gas;
  - ii. pseudo-noble gas; and
  - iii. half-filled orbital.
- c) Describe the formation of the following bonds using Lewis dot symbol:
  - i. ionic or electrovalent bond;
  - ii. covalent bond; and
  - iii. dative or coordinate bond.
- d) Draw Lewis structure of molecules and polyatomic ions with single, double and triple bonds.
- e) Compare the bond length between single, double and triple bonds.
- f) Determine the formal charge and the most plausible Lewis structure.
- g) Explain the exception to the octet rule:
  - i. incomplete octet;
  - ii. expanded octet; and
  - iii. odd number electrons.

**for the cases of molecules with incomplete octet, example $AlCl_3$, $BeCl_2$ - high tendency for the molecules to achieve octet through dimerisation and bridging.*

- h) Illustrate the concept of resonance using appropriate examples.

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4.2 Molecular shape and polarity
- a) Explain Valence Shell Electron Pair Repulsion Theory (VSEPR).
- b) Draw the basic molecular shapes:
  - i. linear;
  - ii. trigonal planar;
  - iii. tetrahedral;
  - iv. trigonal bipyramidal; and
  - v. octahedral.
- c) Predict the shapes of molecule and bond angles in a given species.
- d) Explain bond polarity and dipole moment.
- e) Deduce the polarity of molecules based on the shapes and the resultant dipole moment.

4.3 Orbital overlapping and hybridisation
- a) Illustrate the formation of sigma (σ) and pi (π) bonds from overlapping of orbitals.
- b) Describe the formation of hybrid orbitals of a central atom: sp^3, sp^2, sp, sp^3d, sp^3d^2 using appropriate examples.
- c) Illustrate the hybridisation of the central atom and the overlapping of orbitals in molecules.
  **include examples of molecules with more than one central atom.*

4.4 Intermolecular forces
- a) Describe intermolecular forces:
  - i. van der Waals forces:
    - dipole-dipole interactions or permanent dipole; and
    - London forces or dispersion forces.
  - ii. hydrogen bonding.
- b) Explain factors that influence van der Waals forces and hydrogen bond.
- c) Relate the effects of hydrogen bonding on the following physical properties:
  - i. boiling point;
  - ii. solubility; and
  - iii. density of water compared to ice.

4.5 Metallic bond
- a) Explain the formation of metallic bond by using electron sea model.
- b) Relate metallic bond to the properties of metal:
  - i. malleability;
  - ii. ductility;
  - iii. electrical conductivity; and
  - iv. thermal conductivity.
- c) Explain electrical conductivity of metal by using band theory.
- d) Explain the factors that affect the strength of metallic bond.
- e) Relate boiling/ melting point to the molecular structure, types of bonding and intermolecular forces for elements of:
  - i. period 3;
  - ii. group 1; and
  - iii. group 17.

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Topic 5: States of Matter

5.1 Gas

a) Explain qualitatively the basic assumptions of the kinetic molecular theory of gases for an ideal gas.
b) Define gas laws:
   i. Boyle's law;
   ii. Charles' law; and
   iii. Avogadro's law.
c) Interpret the graphs of Boyle's law and Charles' law.
   **include sketch of graph.*
d) Perform calculations involving gas laws and ideal gas equation.
e) Determine molar mass of a volatile liquid using ideal gas equation.
   (Experiment 4: Charles' law and the ideal gas law).
f) Define Dalton's law.
g) Perform calculations using Dalton's law.
h) Explain qualitatively the ideal and non-ideal behaviours of gases in terms of intermolecular forces and molecular volume.
   **use van der Waals equation to explain the non-ideal behaviour.*
i) Explain the conditions at which real gases approach the ideal behaviour.

5.2 Liquids

a) Relate the properties of liquid to intermolecular forces, molecular arrangement and molecular motion in explaining shape, volume, surface tension, viscosity, compressibility and diffusion.
b) Explain vaporisation and condensation processes based on kinetic molecular theory and intermolecular forces.
c) Define:
   i. vapour pressure; and
   ii. boiling point.
d) Explain boiling process.
e) Illustrate the relationship between:
   i. intermolecular forces and vapour pressure; and
   ii. vapour pressure and temperature.

5.3 Solids

a) Explain the following process using kinetic molecular theory:
   i. freezing (solidification);
   ii. melting (fusion);
   iii. sublimation; and
   iv. deposition.
b) Differentiate between amorphous and crystalline solids.
c) State the following types of crystalline solids with appropriate examples:
   i. metallic;
   ii. ionic;
   iii. molecular covalent; and
   iv. giant covalent.

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5.4 Phase diagram
a) Define:
i. phase;
ii. triple point; and
iii. critical point.
b) Identify triple point and critical point on the phase diagram.
c) Sketch the phase diagram of H_2O and CO_2.
d) Compare the phase diagram of H_2O with CO_2.
e) Explain the anomalous behaviour of H_2O.
f) Describe the changes in phase with respect to:
i. temperature (at constant pressure); and
ii. pressure (at constant temperature).

Topic 6: Chemical Equilibrium

6.1 Dynamic equilibrium
a) Explain the following terms:
i. reversible reaction;
ii. dynamic equilibrium; and
iii. law of mass action.
b) State the characteristics of a system in equilibrium.
c) Interpret the curve of concentration of reactants and products against time for a reversible reaction.
d) Sketch the graph of concentration of reactants and products against time for a reversible reaction.

6.2 Equilibrium constants
a) Define:
i. homogeneous equilibria; and
ii. heterogeneous equilibria.
b) Write expressions for equilibrium constants in terms of concentration, K_c and partial pressure, K_p for homogeneous and heterogeneous systems.
c) Use the equation, $K_p = K_c(RT)^{\Delta n}$ to solve equilibrium problems.
d) Calculate K_c, K_p and the quantities of species present at equilibrium.
e) Determine the degree of dissociation, α.
f) Predict the direction of net reaction by comparing the values of reaction quotient, Q with K.

6.3 Le Chatelier's Principle
a) State Le Chatelier's Principle.
b) Explain the effect of the following factors on a system at equilibrium using Le Chatelier's Principle:
i. concentration of reacting species;
ii. pressure by changing volume;
iii. addition of inert gas at constant volume;
iv. addition of inert gas at constant pressure;
v. temperature; and
vi. catalyst.
(Experiment 5: Chemical Equilibrium)

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Topic 7: Ionic Equilibria

7.1 Acids and bases

a) Define acid and base according to Arrhenius and Bronsted-Lowry theories.
b) Identify conjugate acid and conjugate base according to Bronsted-Lowry theory.
c) Relate pH and pOH to the ionic product of water, K_w at 25 °C.
d) Define strong acid and base, weak acid and base, pH, pOH and pK_w.
e) Calculate the pH values of a strong acid and base.
f) Relate the strength of a weak acid and a weak base to the respective dissociation constants, K_a and K_b.
g) Perform calculations involving:
   i. pH;
   ii. dissociation constant;
   iii. initial concentration;
   iv. equilibrium concentration; and
   v. degree of dissociation, α.
   **perform calculations only on weak monoprotic acids and weak bases.*
h) Explain qualitatively salt hydrolysis using hydrolysis equations of:
   i. strong acid and strong base;
   ii. strong acid and weak base; and
   iii. weak acid and strong base.
i) Classify the salts formed in (h).
j) Define buffer solution.
k) Explain how a buffer solution controls its pH.
l) Write the Henderson-Hasselbalch equation for a given buffer solution.
   **show derivation of Handerson-Hasselbalch equation during T&L process only.*
m) Calculate the pH of buffer solutions using the Henderson-Hasselbalch equation before and after the addition of strong acid or base.

7.2 Acid–base titrations

a) Describe the titration process.
b) Distinguish between the end point and equivalence point.
c) Sketch the titration curves between:
   i. strong acid–strong base;
   ii. strong acid–weak base; and
   iii. weak acid–strong base.
d) Interpret the variation of pH against titre values in (c).
e) Identify suitable indicators for acid-base titrations.
   (Experiment 6: pH measurements and its applications)

7.3 Solubility equilibria

a) Define:
   i. solubility;
   ii. molar solubility; and
   iii. solubility product, K_{sp}.
b) Calculate K_{sp} from concentrations of ions and vice versa.
c) Predict the possibility of precipitation for slightly soluble ionic compounds by comparing the values of ion-product, Q to K_{sp}.
d) Explain the common ion effect.
e) Perform calculations related to common ion effect.

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Panels:

No.	Panel	Institution	Academic Qualification
1.	Prof. Dr. Rosiyah binti Yahya	Universiti Malaya	PhD (Applied Science), Brunel University London, UK BSc (Hons) (Chemistry), Leicester University, UK
2.	Prof. Dr. Zanariah binti Abdullah	Universiti Malaya	PhD (Organic Chemistry), University of London, UK BSc (Hons) (Chemistry), University of East London, UK
3.	Prof. Madya Dr. Che Rozid bin Mamat	Universiti Teknologi Malaysia	PhD (Physical Chemistry), University of Southampton, UK BSc (Hons)(Industrial Chemistry), UTM
4.	Pn. Nor Aziah binti Lazim	Kolej Matrikulasi Selangor	MSc (Chemistry), UKM BSc (Chemistry), Ohio University, USA
5.	Pn. Maizuyah binti Omar	Kolej Matrikulasi Selangor	MSc (HRD & Cognitive Science), UNIMAS BSc(Ed) (Hons) (Chemistry), UKM
6.	Cik Gan Fie Chuen	Kolej Matrikulasi Negeri Sembilan	MEd (Science),UM BSc(Ed) (Hons) (Chemistry), USM
7.	Dr. Azraai bin Othman	Kolej Matrikulasi Kelantan	PhD (Curriculum & Instruction), UPM M.Env (Environmental Science), UPM BSc (Hons) (Chemistry), UKM DipEd (Chemistry), IPGKPP
8.	Pn. Norasyikin binti Ismail @ Chik	Kolej Matrikulasi Kedah	MEd (Chemistry), UPSI BSc(Ed) (Hons) (Chemistry), USM
9.	Pn. Yusmawati binti Yusoff	Kolej Matrikulasi Kejuruteraan Kedah	BSc(Ed) (Hons), USM
10.	Pn. Au Siew Wai	Kolej Matrikulasi Kejuruteraan Johor	BSc(Ed) (Hons) (Chemistry), UKM
11.	En. Mohd Zukiman bin Ismail	Kolej Matrikulasi Kejuruteraan Pahang	BEng (Hons) (Chemical Engineering), USM DipEd (Chemistry), IPGKSM
12.	Pn. Siti Warda binti Selamat	Bahagian Matrikulasi KPM	BScEd (Hons) (Chemistry), UPSI

CURRICULUM SPECIFICATIONS
CHEMISTRY 1 (SK015 & EC015)

CONSTRUCTIVE ALIGNMENT

Course Learning Outcome (CLO)	Programme Learning Outcome (PLO)				Teaching and Learning Method	SLT			%	Assessment	SLT			%	Fixed Assessment Weightage (%)
	PLO 1	PLO 2	PLO 3	PLO 7		F2F	NF2F	Total			F2F	NF2F	Total		
1. Explain the fundamental concepts in chemistry. (C2, PLO 1, MQF LOC i)	✓				Lecture	18	18	36	20	Summative Assessment Test	1.5	4.5	6	24	20
2. Solve problems with analytical and critical thinking using chemistry facts and principles. (C4, PLO 2, MQF LOC ii)		✓			Tutorial (Exam)	46	46	92	51	Final Examination	2	6	8	32	40
					Tutorial (Topic 3)	8	8	16	9	Assignment (CTPS)	0	2	2	8	10
					Lab Discussion	12	12	24	13	Lab Report	1	3	4	16	15
3. Apply the appropriate scientific laboratory skills in chemistry experiments. (P3, PLO 3, MQF LOC iii a)			✓		Practical	12	0	12	7	Practical Test	1	3	4	16	10
4. Analyse numerical problems in chemistry. (C4, PLO 7, MQF LOC iii e)				✓	Tutorial (Topic 3)					Assignment (Numeracy Skill)	0	1	1	4	5
Overall	✓	✓	✓	✓	Lecture Tutorial Practical Lab Discussion	96	84	180	100	Summative Assessment Test Final Examination Assignment Lab Report Practical Test	5.5	19.5	25	100	100

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JADUAL PENENTU UJIAN (JPU)
UJIAN PENILAIAN SUMATIF PROGRAM MATRIKULASI KPM
SISTEM DUA SEMESTER

Kursus: CHEMISTRY 1 **Kod Kursus: SK015 & EC015**

	Topic	CLO	SLT	%	Number of Questions	Taxanomy Level		UPS 1	UPS 2	UPS 3
						C1	C2			
1	Matter	1	4	13.3%	7	/	/	/		
2	Atomic Structure	1	6	20.0%	10	/	/	/		
3	Periodic Table	1	4	13.3%	6	/	/		/	
4	Chemical Bonding	1	6	20.0%	10	/	/		/	
5	States of Matter	1	6	20.0%	10	/	/			/
6	Chemical Equilibrium	1	4	13.3%	7	/	/			/
7	Ionic Equilibria									
TOTAL			**30**	**100.0%**	**50**			**20%**		

Berkuat kuasa mulai sesi 2022/2023

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